PUBLIC PORTFOLIO TRANSLATION

Damavand Predictive Model

Model Selection and Methodology Q&A

Document type: English translation of redacted Greek technical Q&A documentation.

Project context: Original Damavand industrial energy baseline modeling and Measurement & Verification case study.

Use: Public portfolio documentation for the Damavand Energy Forecasting Platform repository.

> **Note.** This is an English translation prepared for public portfolio review. The original Greek redacted documentation remains the authoritative case-study evidence. Sensitive operational data and private details are not included.

Damavand Predictive Model - Model Selection and Methodology Q&A

> English translation prepared for the public portfolio repository.
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> This document is a translated version of the Greek technical Q&A document. It explains the model-selection logic, hyperparameter choices, multicollinearity review, model performance, and robustness checks for the original Damavand M&V baseline model.

1. Selection of the model training method

During development of the baseline model, alternative regression and machine-learning methods were evaluated.

An MLP regressor was initially tested, but it was not selected because it showed lower stability and was less suitable for tabular operational data with strongly correlated variables.

An XGBoost approach was also evaluated. Although it showed extremely strong fit during the training period, with very low training error, this indicated increased flexibility and a higher risk of overfitting relative to the requirements of Measurement & Verification (M&V). In M&V, the priority is not simply maximum predictive accuracy, but a stable and representative reconstruction of pre-intervention behavior.

For this reason, the CatBoost approach was selected. CatBoost is robust to correlated variables, showed stable baseline behavior, and was suitable for reliable energy-savings estimation in line with IPMVP principles.

A similar methodology has also been applied successfully in comparable company projects, producing consistent and technically reliable results.

2. Hyperparameter setting

The final CatBoost model was parameterized with the aim of balancing predictive accuracy and baseline stability, as required in Measurement & Verification applications.

The hyperparameters were selected within recommended ranges from the relevant literature and the official algorithm documentation. They were verified through model behavior during the baseline period and the reporting period.

Specifically, the model used limited tree depth, a controlled learning rate, and a finite number of iterations. This was done to avoid excessive model flexibility and to ensure representative modeling of pre-intervention energy behavior.

The MAE loss function was selected because it is less sensitive to extreme values and aligns with the objective of minimizing total-consumption deviation.

The final hyperparameter configuration was not intended to maximize pure predictive accuracy. Its purpose was to ensure a stable and representative baseline model, as required by M&V practice.

3. Multicollinearity check and explanation

Following a related question, the issue of multicollinearity among independent variables was examined.

Multicollinearity mainly affects linear regression models, where it can distort coefficient estimation. In this study, however, the selected model was a gradient-boosted decision-tree algorithm: CatBoost.

CatBoost is inherently robust to correlated variables because it is based on sequential decision-tree splits rather than linear coefficient estimation. Therefore, correlation between variables does not negatively affect the model's predictive ability or the stability of the baseline model.

For completeness and transparency, a multicollinearity check was still performed, confirming that the selected methodology is appropriate for this dataset.

The findings and selected approach are aligned with the international literature, which recognizes that decision-tree and gradient-boosting models are generally robust to multicollinearity with respect to predictive performance and model stability.

Indicative literature

- Prokhorenkova et al., 2018 - CatBoost
  "Decision tree-based models are generally robust to multicollinearity, as splits are selected based on information gain rather than coefficient estimation."
- Hastie, Tibshirani, Friedman - The Elements of Statistical Learning
  Chapter on Trees & Boosting: multicollinearity primarily affects linear models; tree-based methods are far less sensitive.
- Kuhn & Johnson - Applied Predictive Modeling
  Tree-based models are generally unaffected by multicollinearity in terms of predictive performance.
- Geron - Hands-On Machine Learning
  Correlated features are not a problem for tree-based models; they may affect feature-importance interpretation, but not necessarily accuracy.

VIF results

Feature	VIF
total_kg	595803.1
total_nominal_kg	595437.7
total_brix_units	74.85068
total_hours	1.534043
total_pallets	2.43943
orders	4.269511
avg_brix	4.258338
yield_ratio_actual_over_nominal	16.46171
weekday	2.743861
is_weekend	2.891423
month	123.8769
year	3.118478
week_of_year	124.483

Tables 1 and 2 present the results of the correlation-matrix and Variance Inflation Factor (VIF) checks among the independent variables.

As expected for operational and calendar data from production facilities, a high degree of correlation appears between variables describing related physical or operational quantities.

According to the international literature, multicollinearity is primarily a concern for linear regression models. It does not negatively affect the predictive ability or stability of tree-based and gradient-boosting models such as CatBoost.

For this reason, and given the nature of the selected algorithm, the correlated variables were intentionally retained in the model. They provide a more complete and realistic representation of the facility’s operational and production behavior.

The inclusion of these variables contributes to a realistic description of actual operating conditions and strengthens the representativeness of the baseline model.

The correlation and VIF tables are included for completeness, transparency, and methodological documentation. They confirm the suitability of the selected approach for this dataset.

4. Performance and statistical evaluation of the baseline model

During the baseline period, the model produced a low Mean Absolute Error (MAE = 544.67 kWh) and very limited total deviation (-0.46%).

These results indicate that the model represents the pre-intervention energy behavior of the facility accurately and without systematic bias.

The coefficient of determination, $R^2 = 0.994$, confirms that most of the observed variation in consumption during the baseline period is explained by the model. This is expected for operational data strongly related to production activity.

During the validation period (28/11/2024 - 05/12/2024), which was not included in the training set, the model maintained limited total deviation (-2.24%). This confirms its stability and transferability to a time period outside training.

The negative deviation indicates a tendency to underpredict consumption when the model deviates. In an M&V context, this makes the baseline conservative, because it reduces the risk of overestimating savings.

During the reporting / savings period, the model produced MAE = 5,303.39 kWh. The difference between the estimated baseline consumption and the actual measured consumption corresponds to an estimated energy saving of 11.89%.

The reporting-period $R^2 = 0.925$ indicates that, despite the change in consumption level due to the interventions, the structure and variability of energy behavior are still sufficiently described by the baseline model.

The observed difference between baseline and actual consumption is attributed to real energy-performance improvement, not to loss of predictive capability or model instability.

Variance analysis confirms these findings. During the baseline period, unexplained variance (SSE) is a very small portion of total variance (SST), while most of the variance is explained by the model (SSR).

During the reporting period, residual variance increases, which is expected when operational behavior changes after an intervention. Nevertheless, most of the total variance remains explained by the model, confirming preservation of structural validity.

Overall, the combination of strong baseline fit, conservative validation behavior, and significant positive reporting-period deviation supports the existence of real and measurable energy savings, in line with the logic and principles of IPMVP Option C.

5. Further analysis and robustness of the model to multicollinearity

Although the previous sections document that the selected CatBoost methodology is inherently robust to multicollinearity and does not require removal of correlated variables to ensure predictive ability or baseline stability, an additional sensitivity analysis was performed following a related request.

The purpose was to examine the effect of gradually removing selected correlated variables from the feature set, confirming whether the model results remain consistent and stable.

This analysis was not intended to improve model performance. Its purpose was transparency and additional documentation of the suitability of the initial methodological choice.

Base model, without removing variables

Using the full feature set, the model showed very strong fit during the baseline period ($R^2 = 0.994$) and extremely low total deviation (-0.46%). This indicates accurate, stable, and unbiased representation of pre-intervention energy behavior.

During the reporting period, comparison of estimated baseline consumption with actual measurements showed estimated energy savings of 11.89%, with a high coefficient of determination (R^2 = 0.925).

This confirms that the model does not lose structural validity after the interventions. It continues to describe the structure and variability of consumption adequately.

The observed savings are therefore attributed to real reduction in energy consumption, not to loss of predictive ability or instability of the baseline model.

Removing the variable `total_nominal_kg`

After removing this variable, a small reduction in baseline fit was observed (R^2 = 0.992), with slightly increased error.

However, the estimated savings during the reporting period remained practically unchanged (12.04%). This confirms that the presence or absence of this variable does not materially alter the savings signal.

Removing the variable `week_of_year`

Further removal of the `week_of_year` variable led to a clear increase in estimated savings (14.48%). However, this was accompanied by a significant reduction in reporting-period fit (R^2 = 0.8577) and increased residual variance.

This indicates that although the model produced a higher savings estimate, it lost part of its ability to represent the time and seasonal structure of consumption in a stable way.

That behavior is undesirable for Measurement & Verification applications.

Removing the variable `total_brix_units`

In the final sensitivity scenario, after also removing `total_brix_units`, multicollinearity was reduced substantially, with all VIF values below 5.

During the baseline period, the model maintained good fit, with MAE = 523.09 kWh and total deviation = -0.65%. This indicates acceptable and unbiased representation of pre-intervention consumption.

During validation, deviation was -3.72% with MAE = 1,661.79 kWh. This indicates that, even when the model deviates from measured values, it tends to underpredict energy consumption.

In Measurement & Verification applications, this is conservative because it reduces the risk of overestimating savings.

During the reporting period, however, model stability decreased. The model showed lower R^2 (0.898) and increased error variability (MAE = 5,335.25 kWh) compared with the base scenario.

The estimated energy saving was 11.76%, comparable to the full model. However, there was no clear advantage in terms of overall stability or baseline reliability.

Therefore, although removing this variable further reduces multicollinearity, it does not materially improve performance or reporting-period behavior. This confirms the suitability of retaining the full feature set in the final scenario.

Overall evaluation

The sensitivity analysis shows that removing correlated variables does not lead to meaningful improvement in the stability or reliability of the baseline model.

In some cases, the estimated savings percentage changes, but those scenarios are also accompanied by reduced ability to represent the facility's full operational and production behavior during the reporting period.

For this reason, although removing variables could have led to marginally higher estimated savings in some scenarios, the full feature set was intentionally retained in the final model. This decision was not based only on predictive accuracy, but on the need for a more complete, realistic, and physically interpretable representation of the facility, using the full set of available production and operational data.

This approach is fully aligned with the principles of IPMVP Option C, where the priority is baseline stability and representativeness rather than simply pursuing changes in the estimated savings magnitude.

6. References

CatBoost official literature

1. Prokhorenkova et al. (2018)
   CatBoost: unbiased boosting with categorical features, NeurIPS 2018.
   Describes regularization, ordered boosting, and robustness.
2. CatBoost Documentation - Parameter Tuning Guide
   https://catboost.ai/docs/concepts/parameter-tuning.html
   Recommended ranges for depth, learning rate, and iterations.

Gradient boosting and overfitting

1. Friedman, J. (2001)
   Greedy Function Approximation: A Gradient Boosting Machine, Annals of Statistics.
   Foundational paper for boosting and the trade-off between learning rate and trees.
2. Hastie, Tibshirani, Friedman (2009)
   The Elements of Statistical Learning, Springer.
   Chapter on boosting: regularization and model complexity.

Energy modeling / M&V context

1. ASHRAE Guideline 14
   Measurement of Energy, Demand, and Water Savings
   Recommends stability and suitable error metrics, not a perfect fit.
2. IPMVP Core Concepts (EVO)
   Emphasizes baseline representativeness, conservative modeling, and avoidance of bias.

End of English translation.